

InvisiGuard - Intelligent System for Predicting Cyber Threats in Real Time

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Declaration

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ABSTRACT

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List of abbreviations

Abbreviation	Full Form
API	Application Programming Interface
ATT&CK	Adversarial Tactics, Techniques, and Common Knowledge
AES	Advanced Encryption Standard
BRIN	Block Range Index
CPU	Central Processing Unit
CSV	Comma-Separated Values
DLL	Dynamic Link Library
EDR	Endpoint Detection and Response
FPR	False Positive Rate
GDPR	General Data Protection Regulation
GIN	Generalized Inverted Index
HA	High Availability
IDS	Intrusion Detection System
JSON	JavaScript Object Notation
JSONB	JSON Binary
JWT	JSON Web Token
LSASS	Local Security Authority Subsystem Service
MITRE	Massachusetts Institute of Technology Research and Engineering
ML	Machine Learning
MUI	Material-UI
ORM	Object-Relational Mapping

Abbreviation	Full Form
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
REST	Representational State Transfer
SIEM	Security Information and Event Management
SOAR	Security Orchestration, Automation and Response
SQL	Structured Query Language
SSE	Server-Sent Events
SVM	Support Vector Machine
TF-IDF	Term Frequency-Inverse Document Frequency
TLS	Transport Layer Security
UUID	Universally Unique Identifier
VM	Virtual Machine
WEF	Windows Event Forwarding
XGBoost	Extreme Gradient Boosting

1.INTRODUCTION

The exponential growth of cyber threats targeting enterprise endpoints has created an urgent need for intelligent, real-time security monitoring solutions. Traditional signature-based antivirus systems and rule-based intrusion detection systems (IDS) are increasingly inadequate against sophisticated, polymorphic malware and advanced persistent threats (APTs). Modern adversaries employ living-off-the-land (LotL) techniques, leveraging legitimate system tools like PowerShell, WMI, and native Windows utilities to evade detection while executing malicious activities.

InvisiGuard addresses this critical security challenge by implementing a machine learning-powered endpoint detection and response (EDR) platform that analyzes Windows event logs in real-time. The system ingests telemetry from Windows endpoints instrumented with Sysmon, applies advanced feature engineering to extract 369 features, and employs an XGBoost multi-class classifier to identify malicious activities with high accuracy and low false positive rates.

This documentation presents a comprehensive technical analysis of the InvisiGuard platform, examining its architectural design, machine learning methodology, implementation details, and operational characteristics. The system represents a convergence of cybersecurity domain knowledge, machine learning engineering, and distributed systems design to create a production-ready threat detection solution.

1.1 Background Literature

Evolution of Endpoint Security

The landscape of endpoint security has undergone significant transformation over the past three decades. Traditional antivirus solutions, introduced in the 1990s, relied primarily on signature-based detection methods that compared file hashes and byte patterns against known malware databases. While effective against known threats, this approach proved fundamentally reactive and ineffective against zero-day exploits and polymorphic malware that could alter their signatures with each infection.

The mid-2000s witnessed the emergence of heuristic analysis and behavioral detection systems that attempted to identify malicious activity based on program behavior rather than static signatures. These systems monitored process execution patterns, file system modifications, registry changes, and network communications to detect anomalous behavior indicative of malware. However, early behavioral systems suffered from high false positive rates and limited visibility into sophisticated attack techniques.

The concept of Endpoint Detection and Response (EDR) emerged around 2013, coined by Anton Chuvakin of Gartner, representing a paradigm shift from prevention-focused to detection-and-response-focused security. EDR solutions emphasized continuous monitoring, data collection, threat detection, investigation capabilities, and remediation guidance. Modern EDR platforms collect extensive telemetry from endpoints, including process creation events, network connections, file operations, registry modifications, and memory access patterns.

Machine Learning in Cybersecurity

The application of machine learning to cybersecurity problems has accelerated dramatically since 2015, driven by advances in computational power, availability of large labeled datasets, and improvements in algorithm efficiency. Several research directions have emerged:

Supervised Learning for Malware Classification:

Researchers have extensively explored supervised learning algorithms for binary classification (benign vs. malicious) and multi-class classification (identifying specific malware families or attack types). Saxe and Berlin (2015) demonstrated that deep neural networks could achieve 95% detection accuracy on portable executable (PE) files using static analysis features. Raff et al. (2018) showed that convolutional neural networks could learn effective representations directly from raw byte sequences.

Anomaly Detection for Intrusion Detection:

Unsupervised and semi-supervised learning approaches have been applied to identify anomalous behavior in network traffic and system logs. Chandola et al. (2009) provided a comprehensive survey of anomaly detection techniques, categorizing approaches into statistical methods, clustering-based methods, and neural network-based methods. One-Class SVM, introduced by Schölkopf et al. (2001), has proven particularly effective for novelty detection in security contexts where labeled malicious samples are scarce.

Ensemble Methods and Gradient Boosting:

Ensemble learning methods, particularly gradient boosting algorithms, have demonstrated superior performance in cybersecurity applications. Chen and Guestrin (2016) introduced XGBoost (Extreme Gradient Boosting), which has become the algorithm of choice for many security researchers due to its speed, accuracy, and ability

to handle imbalanced datasets. XGBoost has been successfully applied to malware detection (Azeez et al., 2020), intrusion detection (Dhaliwal et al., 2018), and phishing detection (Somesha et al., 2020).

Windows Event Log Analysis

Windows operating systems generate extensive telemetry through the Event Tracing for Windows (ETW) framework, which provides a unified mechanism for kernel and user-mode components to log events. The Windows Event Log service records these events in structured XML format, categorized into system logs, application logs, and security logs.

Sysmon (System Monitor):
Sysmon, developed by Mark Russinovich and Thomas Garnier at Microsoft Sysinternals, extends Windows logging capabilities by recording detailed information about process creation, network connections, file creation time changes, and other system activities. Sysmon has become the de facto standard for endpoint telemetry collection in security operations centers (SOCs) due to its granular visibility and configurable event filtering.

Research by Bohannon and Holmes (2017) demonstrated that Sysmon logs contain sufficient information to detect advanced attack techniques including process injection, credential dumping, and lateral movement. The MITRE ATT&CK framework, introduced by MITRE Corporation in 2013, provides a comprehensive knowledge base of adversary tactics and techniques, many of which can be detected through analysis of Sysmon event logs.

Event Log Analysis Techniques:
Several research efforts have focused on automated analysis of Windows event logs for security purposes. Kent et al. (2015) proposed using process tree reconstruction from Sysmon Event ID 1 (Process Creation) to identify malicious process chains. Hossain et al. (2017) demonstrated that TF-IDF (Term Frequency-Inverse Document Frequency) vectorization of command-line arguments could effectively distinguish benign administrative activity from malicious PowerShell execution.

Feature Engineering for Endpoint Security

Effective machine learning for cybersecurity requires domain-informed feature engineering that captures behavioral indicators of malicious activity. Research has identified several categories of discriminative features. InvisiGuard specifically implements the following three categories based on Sysmon event log data:

Process-Based Features:

- Process name and path analysis (legitimate vs. suspicious locations)
- Parent-child process relationships (unusual spawning patterns)
- Command-line argument analysis (encoded commands, suspicious flags)
- Process integrity levels and privilege escalation indicators

File System Features:

- File creation, modification, and deletion patterns
- Access to sensitive directories (System32, user profiles)
- Executable file characteristics (signed vs. unsigned, hash reputation)
- File extension mismatches and double extensions

Registry Features:

- Modifications to autorun locations (persistence mechanisms)
- Changes to security settings and policies
- Creation of suspicious registry keys
- Access to credential storage locations

Real-Time Stream Processing

Modern security platforms require real-time processing of high-volume event streams to provide timely threat detection and response. Apache Kafka, introduced by LinkedIn in 2011 and open-sourced in 2012, has emerged as the leading distributed streaming platform for security telemetry ingestion. Kafka provides fault-tolerant, high-throughput message queuing with persistent storage, enabling reliable log collection even during system failures or maintenance windows.

Research by Kreps et al. (2011) established the architectural principles of log-centric data integration, demonstrating that treating all data as an append-only log enables scalable, fault-tolerant distributed systems. This approach has been widely adopted in security operations, where log data serves as the authoritative record of system activity.

MITRE ATT&CK Framework

The MITRE ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge) framework, first released in 2013 and continuously updated, provides a comprehensive taxonomy of adversary behavior based on real-world observations. The framework organizes attack techniques into tactical categories including Initial Access, Execution, Persistence, Privilege Escalation, Defense Evasion, Credential Access, Discovery, Lateral Movement, Collection, Command and Control, Exfiltration, and Impact.

ATT&CK has become the industry standard for describing and categorizing cyber threats, enabling security teams to communicate about adversary behavior using common terminology. Modern EDR platforms map their detections to ATT&CK techniques, providing context about the stage of an attack and potential adversary objectives.

Research by Strom et al. (2018) demonstrated that ATT&CK-based threat modeling significantly improves security team effectiveness by providing a structured approach to understanding adversary behavior and prioritizing defensive investments. The framework's technique descriptions include detection guidance, specifying which data sources and analytics can identify each technique.